Table of Contents

Plotting the graphs of polynomials of degree 4 and 5, their 1

Plotting the graphs of polynomials of degree 4 and 5, their

```
%%first and second derivatives and analysis of these graphs.

%Shushank

close all;
clc;

% Define a range of x values
x = linspace(-2, 2, 1000);

% Define coefficients for the degree 4 and 5 polynomials
coefficients_deg4 = [1, -3, 2, 0, 1];
coefficients_deg5 = [0.5, 1, -2, -1, 2, 1];

% Evaluate the polynomials and their derivatives
poly_deg4 = polyval(coefficients_deg4, x);
poly_deg5 = polyval(coefficients_deg5, x);

% Calculate first and second derivatives
poly_deg4_derivative1 = polyder(coefficients_deg4);
poly_deg5_derivative1 = polyder(coefficients_deg5);

poly_deg4_derivative2 = polyder(poly_deg4_derivative1);
poly_deg5_derivative2 = polyder(poly_deg5_derivative1);

% Create graph for Degree 4 Polynomial
figure(1);
plot(x, poly_deg4);
title('Degree 4 Polynomial');
xlabel('x');
ylabel('y');
grid on;

% Create graph for Degree 5 Polynomial
figure(2);
plot(x, poly_deg5);
title('Degree 5 Polynomial');
xlabel('x');
ylabel('y');
grid on;

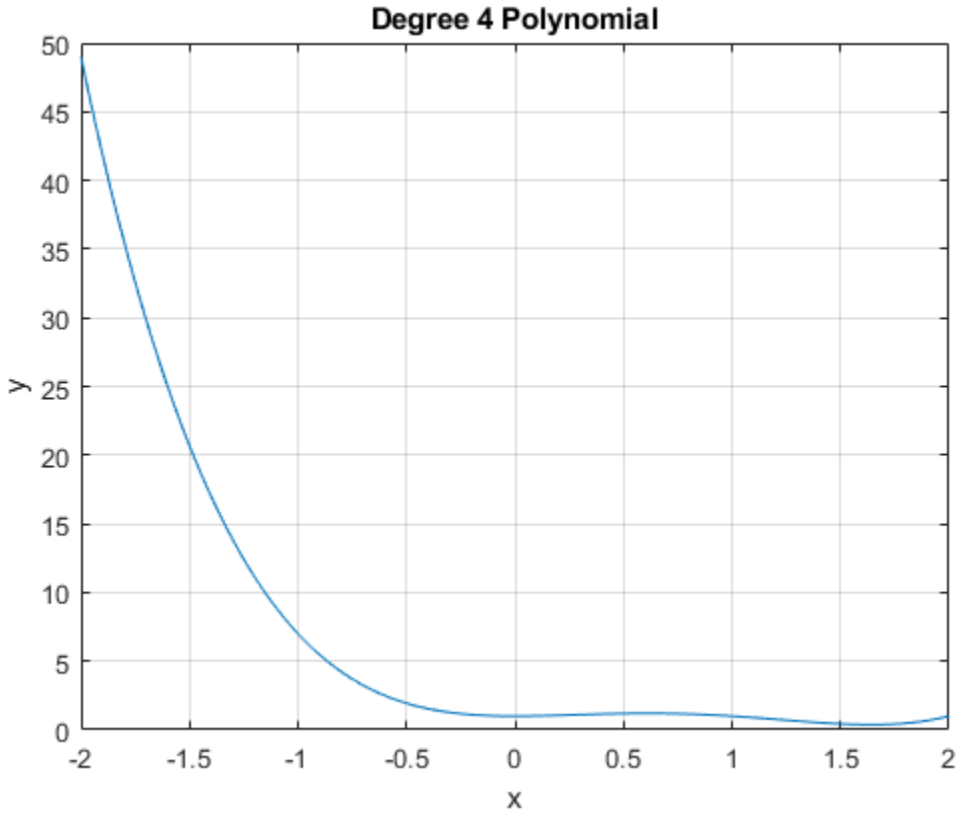
% Create graph for First Derivative of Degree 4 Polynomial
figure(3);
```

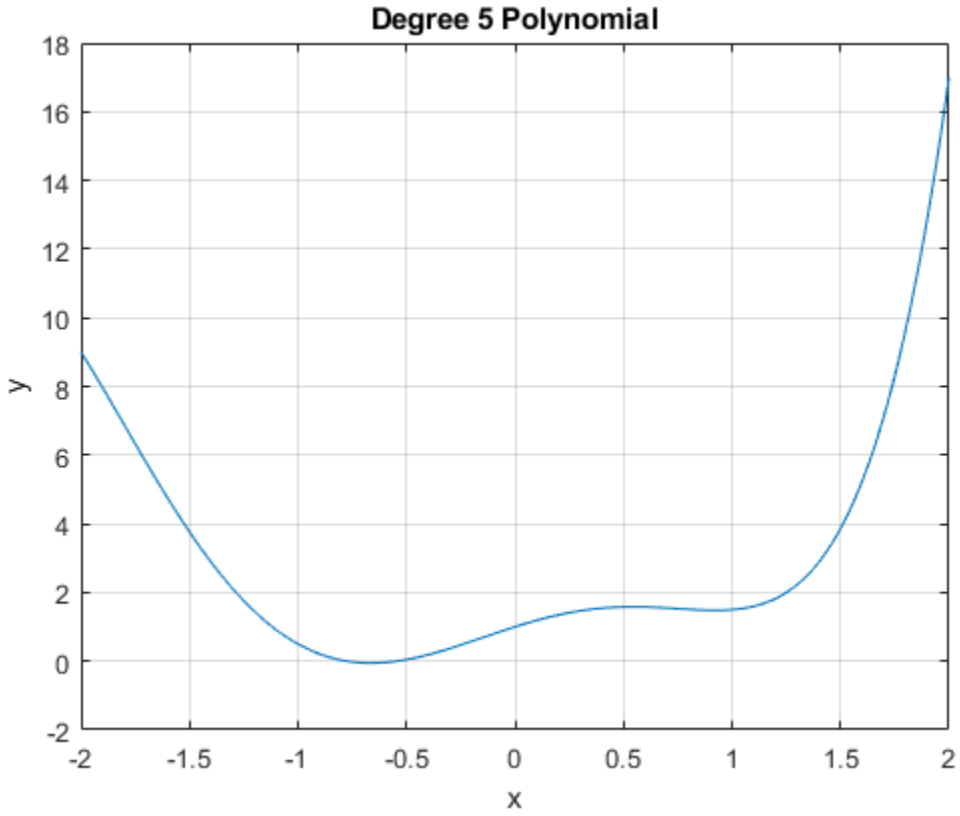
```
plot(x, polyval(poly_deg4_derivative1, x));
title('First Derivative of Degree 4 Polynomial');
xlabel('x');
ylabel('y');
grid on;

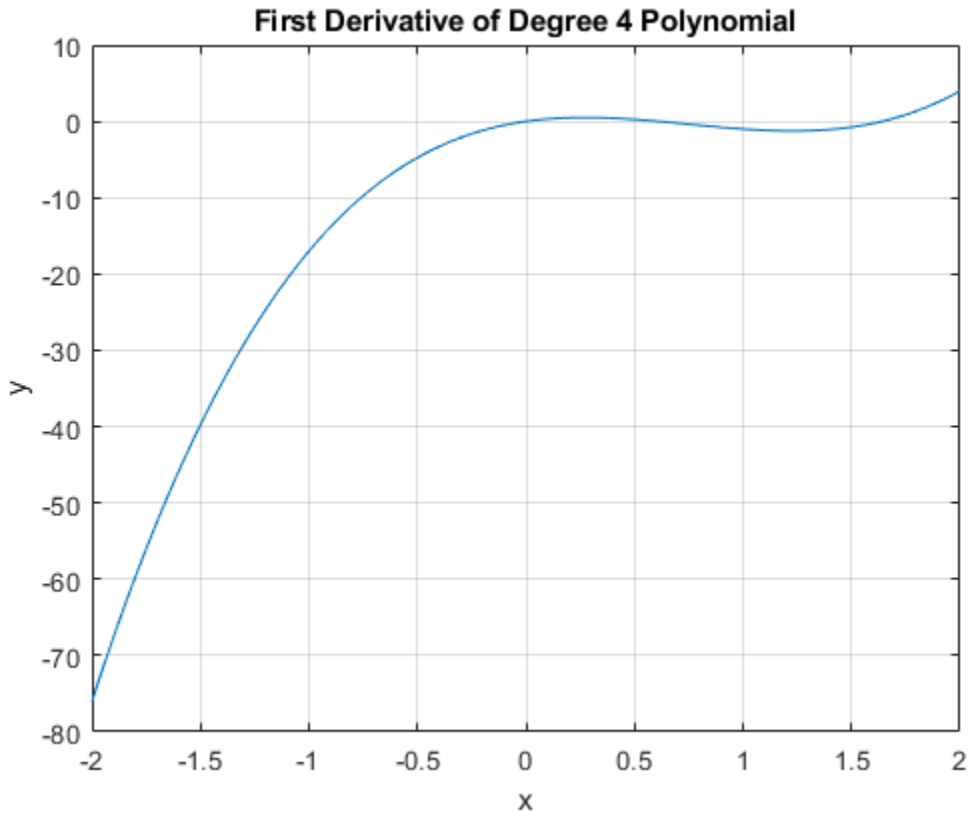
% Create graph for First Derivative of Degree 5 Polynomial
figure(4);
plot(x, polyval(poly_deg5_derivative1, x));
title('First Derivative of Degree 5 Polynomial');
xlabel('x');
ylabel('y');
grid on;

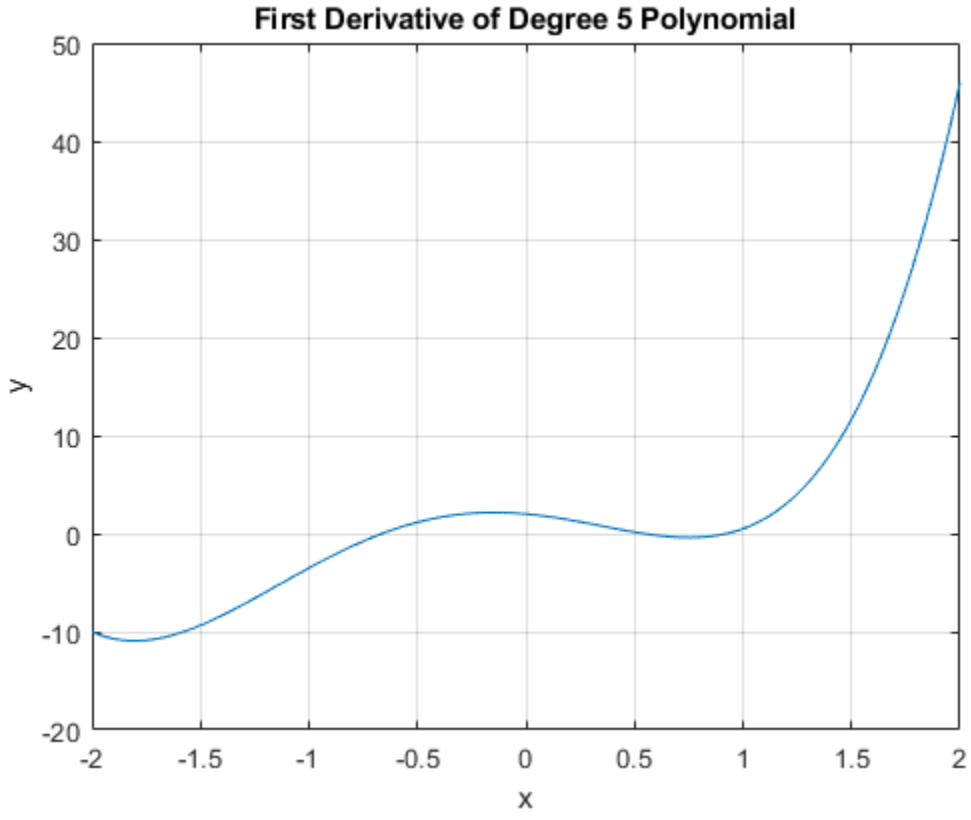
% Create graph for Second Derivative of Degree 4 Polynomial
figure(5);
plot(x, polyval(poly_deg4_derivative2, x));
title('Second Derivative of Degree 4 Polynomial');
xlabel('x');
ylabel('y');
grid on;

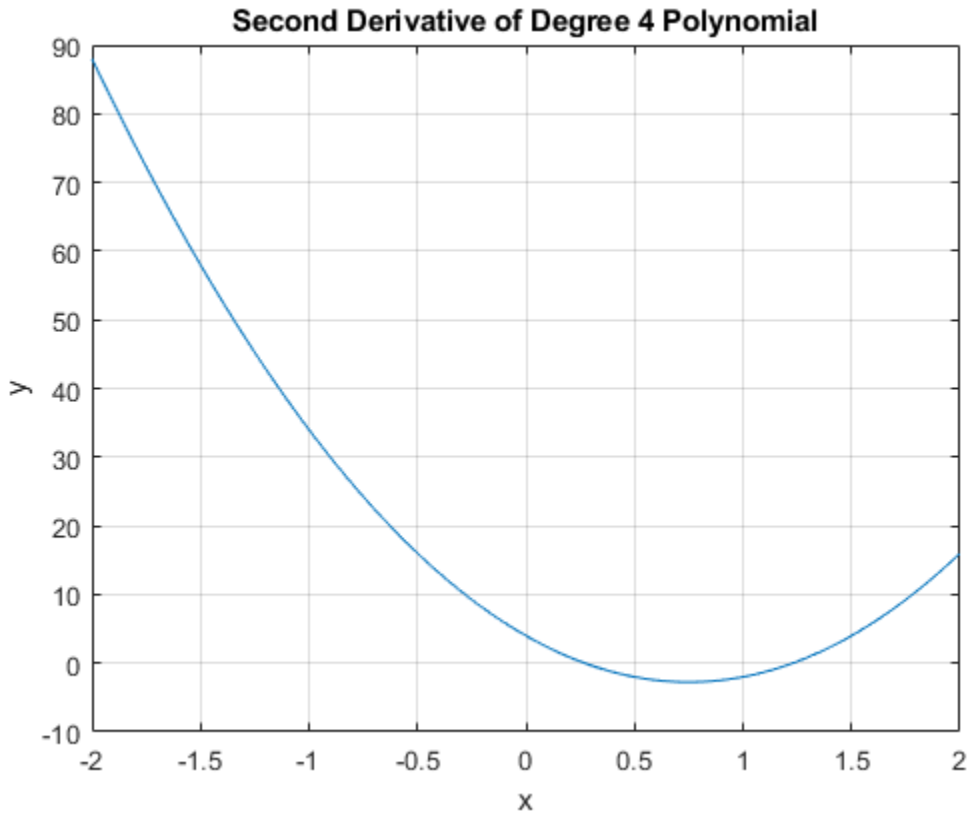
% Create graph for Second Derivative of Degree 5 Polynomial
figure(6);
plot(x, polyval(poly_deg5_derivative2, x));
title('Second Derivative of Degree 5 Polynomial');
xlabel('x');
ylabel('y');
grid on;
```

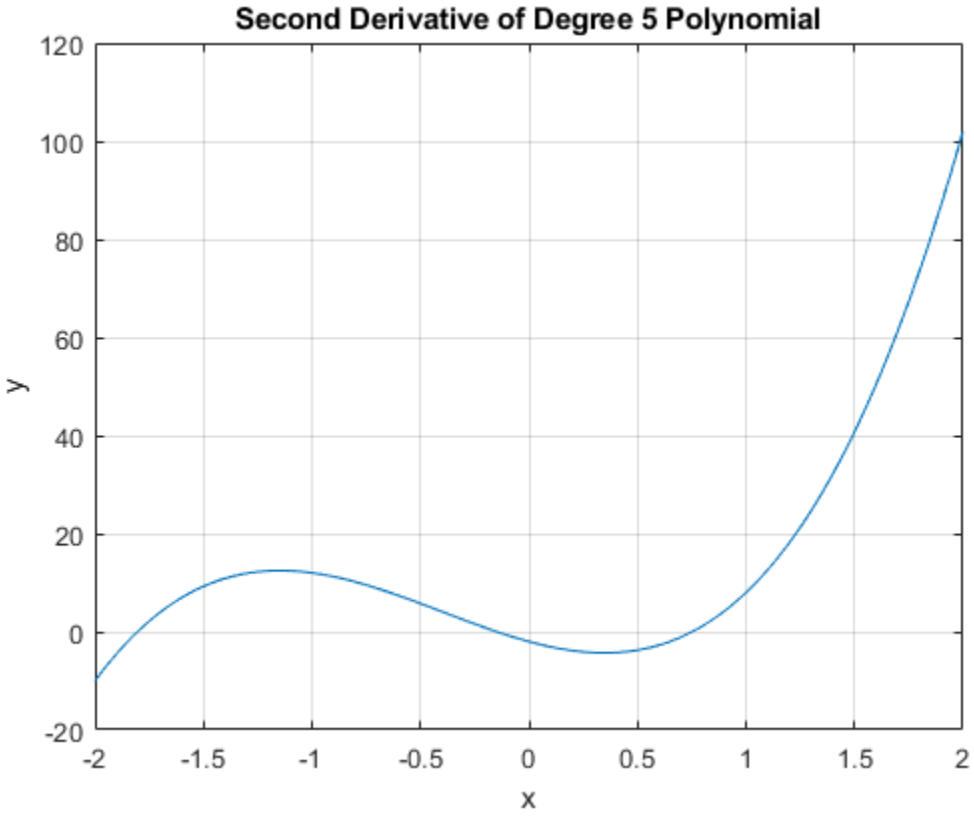

Degree 4 Polynomial
50
45
40
35
30
25
20
15
10
5
0
y
-2
-1.5
-1
-0.5
0
0.5
1
1.5
2
x


Degree 5 Polynomial
18
16
14
12
10
8
6
4
2
0
-2
y
-2
-1.5
-1
-0.5
0
0.5
1
1.5
2
x


First Derivative of Degree 4 Polynomial
10
0
-10
-20
-30
-40
-50
-60
-70
-80
y
-2
-1.5
-1
-0.5
0
0.5
1
1.5
2
x


First Derivative of Degree 5 Polynomial
50
40
30
20
10
0
-10
-20
y
-2
-1.5
-1
-0.5
0
0.5
1
1.5
2
x


Second Derivative of Degree 4 Polynomial
90
80
70
60
50
40
30
20
10
0
-10
y
-2
-1.5
-1
-0.5
0
0.5
1
1.5
2
x


Second Derivative of Degree 5 Polynomial
120
100
80
60
40
20
0
-20
y
-2
-1.5
-1
-0.5
0
0.5
1
1.5
2
x

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